

LAB 3: Propositional Logic & Model Checking

Group Challenge

Instructor	Dr. Naila Marir	Duration	1 hour 30 minutes
Semester	Fall 2026	Group size	3–4 students
Submission	One GitHub link per group	Total marks	100
Group No.			
Member name	Student ID	Member name	Student ID

1. Objectives

1. Represent facts as propositional logic sentences using Not, And, Or and Implication.
2. Explain how `model_check` decides whether a knowledge base entails a query.
3. Encode a new logic puzzle in Python and solve it with model checking.

2. Tools and Files

1. Python 3 and the package `termcolor` (install with `pip install termcolor`).
2. Lab files (on Blackboard): `logic.py`, `harry.py`, `clue.py`, `puzzle.py`, `mastermind.py`. Put all of them in one folder.
3. A GitHub account for each group.

3. Lab Plan

Step	Activity	Time
0	Setup: run <code>harry.py</code> and <code>clue.py</code>	5 min
1	Task 1 — Understand the code	20 min
2	Task 2 — Change the knowledge	20 min
3	Task 3 — Knights & Knaves challenge	30 min
4	Write the PDF and push to GitHub	15 min

4. Task 1 — Understand the Code (30 marks, 20 min)

1. **harry.py:** Write the truth table for `rain`, `hagrid`, `dumbledore` (8 rows). Mark the rows where the knowledge base is true. Use the table to explain why the program prints `True`.

Harry.py Truth table

P: it is raining Q: harry will visit hagrid R: harry will visit dumbledore

$\rightarrow (\neg P \rightarrow Q) \wedge (Q \vee R) \wedge \neg(Q \wedge R) \wedge R$

P	Q	R	$\neg P \rightarrow Q$	$Q \vee R$	$\neg(Q \wedge R)$	R	KB
False	False	False	F	F	T	F	F
False	False	True	T	T	T	T	F
False	True	False	T	T	T	F	F
False	True	True	T	T	F	T	F
True	False	False	F	F	T	F	F
True	False	True	T	T	T	T	<u>T</u>
True	True	False	T	T	T	F	F
True	True	True	T	T	F	T	F

2. **clue.py:** The program prints MsScarlet, library and knife. Explain in plain English why each one must be true.

The output of the code "Msscarlet, library and knife" is true, because with the knowledge of the initial cards being not mustard, not kitchen and not wrench, with the combined knowledge of the known cards being not plum and not ballroom, we are left with only ms scarlet as the person, the library as the room, as for the weapon, and because of the unknown cards we know that one of the cards is not the answer, and since its scarlet , and library then its not the wrench, but the knife.

5. Task 2 — Change the Knowledge (30 marks, 20 min)

For each change: (a) predict the result, (b) run the code, (c) explain the result.

1. **harry.py:** Delete the last fact `dumbledore` from the knowledge base. What does the program print now? Why?

Because we know it's raining because of the knowledge that harry visited dumbledore, without that knowledge, it cannot guarantee that it was actually raining

2. **puzzle.py:** Delete the clue `MinervaGryffindor`. Who can still be placed in a house? Then add **one new clue of your own** that makes the answer unique again.

no one can be placed for sure in any house because glideroy is either in grtffindor or ravenclae ane pomona is not in slytherin, it depends on MinervaGryiffendor clue.

6. Task 3 — Group Challenge: Knights & Knaves (30 marks, 30 min)

On an island, a **knight** always tells the truth and a **knave** always lies. Every person is one or the other, never both. Create a new file `knights.py` that imports `logic.py` and solves the puzzles below.

Hint — use this pattern for what a person says:

```
Implication(AKnight, statement)      # a knight's statement is true
Implication(AKnave, Not(statement))  # a knave's statement is false
```

1. **Puzzle 0:** A says “I am both a knight and a knave.”
2. **Puzzle 1:** A says “We are both knaves.” B says nothing.
3. **Puzzle 2:** A says “We are the same kind.” B says “We are of different kinds.”

Bonus (+10 marks) — Puzzle 3: A says either “I am a knight.” or “I am a knave.” (you don’t know which). B says “A said ‘I am a knave.’” B also says “C is a knave.” C says “A is a knight.”

7. Submission

Create **one public GitHub repository** named `Lab3_GroupX` and submit **only its link** on Blackboard. The repository must contain:

1. `Lab3_GroupX.pdf` — group number, member names, and for each task: your answer, your code, and a screenshot of the output.
2. All `.py` files — `logic.py`, your changed `harry.py` and `puzzle.py`, and your new `knights.py`.
3. `README.md` — member names and one line on how to run `knights.py`.

Before you submit: open your link in a private (incognito) browser window to check that it works.

8. Grading Rubric

Task	What is checked	Marks
Task 1	Correct truth table and clear explanations	30
Task 2	Predictions, results and explanations; a working new clue	30
Task 3	Puzzles 0–2 encoded correctly with output and explanation	30
Submission	Clear PDF and complete GitHub repository	10
Total		100
Bonus	Puzzle 3 solved and explained	+10